BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: DIK-HSSC-XII-2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XII-2022 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. According to special theory of relativity _____

(A) Moving clock tick slowly in comparison to stationary clock
(B) Moving clock tick quickly in comparison to stationary clock
(C) The two clock will tick at the same speed
(D) None

Qii. When the number of electrons in excited state exceeds than the number of electrons in the ground state, it is called _____

(A) Excitation
(B) Ionization
(C) Population inversion
(D) None

Qiii. The energy of the electron in excited state n=4 in H-atom is _____

(A) $-13.6eV$
(B) $-3.4eV$
(C) $-1.5eV$
(D) $-0.85eV$

Qiv. All free particles have a color charge of _____

(A) 0
(B) 2/3
(C) 1/3
(D) 1

Qv. Which of the following is a boson?

(A) Proton
(B) Neutron
(C) Photon
(D) Pion

Qvi. If 15/16 of the radioactive atoms of an elements decay in 8 minutes, the half-life of elements is _____

(A) 8 mint
(B) 4 mint
(C) 2 mint
(D) 12 mint

Qvii. The capacitance of the parallel capacitor having vacuum between the plates in, where K is coulumb constant?

(A) ε_0/Ad
(B) $A/(4\pi Kd)$
(C) $d\varepsilon_0/A$
(D) All

Qviii. A proton is about 1840 time heavier than an electron. When it is accelerated by a P.D of 1KV, its K.E will be:

(A) 1840 KeV
(B) 1/1840 KeV
(C) 1 KeV
(D) 920 KeV

Qix. 10^6 electrons are moving through a wire per second, the current developed is _____

(A) $1.6x10^{-19}A$
(B) 10^6A
(C) 1A
(D) $1.6x10^{-13}A$

Qx. A current of 9.0 A flows in a circuit of three resistors connected in parallel to a P.D source if each resistor has magnitude 5Ω, then current in the 3rd resistor is _____

(A) 9A
(B) 3A
(C) 1.5A
(D) 0.5A

Qxi. One weber is equal to _____

(A) NA^2/m
(B) Nm^2/A
(C) Nm/A
(D) NA/m

Qxii. Set-up transformer has a transformation ratio of 3:2. What is the voltage in secondary of voltage in primary is 30V?

(A) 45V
(B) 15V
(C) 90V
(D) 300V

Qxiii. r.m.s value of a sine die wave is 100A, its peak value is

(A) 70.7A
(B) 141.4A
(C) 150A
(D) 282.2 [illegible in source scan]

Qxiv. A capacitor is perfectly for _____

(A) Direct current
(B) Alternating current
(C) Induced emf
(D) Both A and B

Qxv. A force of 500N is applied to one end of a cylindrical [illegible in source scan] diameter 50cm. The tensile stress in N/m^2 is _____

(A) 500
(B) 2.5×10^4
(C) 2.5×10^{-4}
(D) 2.5 x [illegible in source scan]

Qxvi. Ratio of stress to strain is _____

(A) 0
(B) >1
(C) Constant
(D) <1

Qxvii. In transistor the central region base is doped _____

(A) Heavily
(B) Moderately
(C) Lightly
(D) None

Qxviii. Pair production cannot take place in vacuum because _____

(A) Mass is not conserved
(B) Momentum is not conserved
(C) Energy is not conserved
(D) Charge is not conserved

SECTION – B

Time Allowed: 2:40 Hrs

Q2. Attempt any Ten questions of the following. All questions carry equal marks

i. Water has a large dielectric constant, but it is rarely [illegible in source scan] in capacitor why?

ii. State Ohm's law.

iii. How does a current carrying coil behave like a [illegible in source scan]?

iv. Does a moving charge experiences a force in [illegible in source scan] field. Explain.

v. What is meant by the term back e.m.f in any [illegible in source scan] motor operation?

vi. The voltage across a 0.5 μF capacitor is 16sin(2π [illegible in source scan] Find the capacitive reactance.

vii. Distinguish between soft and hard substance by [illegible in source scan] its curves,

viii. Deduce the relation between α and β of a [illegible in source scan]

ix. Distinguish between N-type and P-type semiconductors.

x. Why must the rest mass of a photon be zero?

xi. Discuss the principle and operation of a laser.

xii. Why are small and large nuclei unstable?

xiii. Explain the term mass defect and binding energy.

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any three Question. Each question carries equal marks.

Q.3 a. Describe the process of charging and discharging a capacitor. Give the diagram and mathematical expression.

b. How a 5mA, 100 ohms galvanometer is converted into 20V voltmeter?

Q.4 a. Explain the phenomenon of self-induction and mutual induction.

b. The current through a 60mH inductor $0.2\sin(377t-25°)$. Write the mathematical expression for voltage across it.

Q.5 a. State Einstein's Postulates of special theory of relativity. Discuss its various results.

b. An electron is in the first Bohr orbit of H-atom. (i). Find the speed of electron (ii). The time required for electron to circle the nucleus.

Q.6 Write notes on any two of the followings. a. Kirchhoff's laws b. R.L series A.C circuit c. Nuclear fission